GHANA COMMUNICATION TECHNOLOGY UNIVERSITY

FACULTY OF COMPUTING AND INFORMATION SYSTEM.

DEPARTMENT OF INFORMATION SYSTEMS.

Title:

Development of an AI-Powered Contract Clause Risk Analyzer for Enhanced Legal Literacy among Freelancers and SMEs

A research proposal submitted as fulfillment of the degree of BSC in Computer Science

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ABSTRACT

High legal costs limit access to professional legal advice for freelancers and small businesses.
This study proposes an automated legal risk detection and simplification system.
A Design Science Research approach is used to develop a web-based prototype.
Natural Language Processing and Explainable AI identify high-risk contractual clauses.
Legal-BERT supports clause classification with a risk scoring mechanism.
The system enhances contractual understanding through an affordable LegalTech solution.

1.1. INTRODUCTION

The legal landscape is increasingly complex, making the review of commercial agreements—such as Non-Disclosure Agreements (NDAs) and Service Agreements—a critical task for any business entity. However, for freelancers, small business owners, and startup founders, the cost of professional legal counsel, which often exceeds $300 per hour, creates a significant barrier to justice. As these unrepresented parties seek to navigate the digital economy, they are frequently forced to "sign blind," exposing themselves to high-risk clauses like indefinite indemnification, broad non-competes, and unfavorable payment terms.

Traditional methods of managing legal risk—relying on human lawyers or basic keyword searches—are often too expensive or inefficient for the high volume of documents generated in modern business. While digital tools have emerged, many lack the specialized intelligence required to interpret the nuances of legal prose. Natural Language Processing (NLP) and Artificial Intelligence (AI) offer a promising solution to this challenge by automating the analysis of unstructured text.

This study introduces an **AI-Powered Contract Clause Risk Analyzer**, a web-based application designed to bridge the legal gap through **Explainable AI (XAI)**. By leveraging **Legal-BERT**, a model pre-trained on legal corpora, the system can classify specific clauses and assess their risk levels. Furthermore, the integration of Large Language Models (LLMs) allows the system to translate complex legalese into plain English, providing users with a clear understanding of their obligations and suggesting neutral alternatives.

The research is grounded in the **Design Science Research (DSR)** methodology, focusing on the iterative development and evaluation of a functional prototype. This project seeks to demonstrate that AI can reliably identify "unfair" terms and provide layman-friendly explanations without compromising legal accuracy. Ultimately, this research contributes to the fields of LegalTech and Human-Computer Interaction by providing an accessible, "access to justice" tool for the modern workforce.

1.2 BACKGROUND TO THE STUDY

The integration of artificial intelligence (AI) into the legal domain has significantly transformed how legal and administrative services are delivered. Traditionally, the interpretation of commercial contracts was a manual, labor-intensive process reserved for legal professionals. However, as the digital economy expands, the demand for timely and efficient contract review has increased, particularly for freelancers and SMEs who lack the resources for traditional legal support. This has led to a growing interest in leveraging AI to improve service delivery and accessibility.

Despite the emergence of basic legal tools, many existing systems remain limited in their ability to handle complex or context-specific queries. Such systems often rely on keyword matching, which fails to capture the nuanced legal implications of unstructured text. To address these limitations, researchers are now incorporating **Explainable AI (XAI)** and specialized **Natural Language Understanding (NLU)** techniques into legal technologies. NLU enables machines to interpret user intent and extract meaning from complex language, which is critical in a domain where a single word can alter a party's liability.

In the context of "Access to Justice," these intelligent systems play a vital role in protecting unrepresented parties. They offer 24/7 assistance, reduce the burden on human advisors, and provide instant information that would otherwise be inaccessible due to high costs. As the professional world becomes more digitized, the demand for reliable, automated tools to interpret institutional policies and legal documents continues to grow.

This study seeks to design and develop an **AI-Powered Contract Clause Risk Analyzer** that leverages specialized NLP models to support users in identifying and understanding legal risks. By using models like **Legal-BERT** and datasets like **CUAD**, the proposed system aims to overcome the shortcomings of traditional keyword-based tools and contribute to the broader application of AI in promoting legal equity.

1.3 PROBLEM STATEMENT

As the gig economy and startup ecosystem expand, the volume of commercial agreements—such as service contracts and NDAs—has increased significantly. However, traditional legal support systems are often unable to handle the needs of freelancers and small business owners efficiently due to the high cost of legal consultation, which frequently exceeds $300 per hour. This financial barrier creates an "Access to Justice" gap, where unrepresented parties are forced to sign complex legal documents without a full understanding of the risks involved.

While some digital tools have been adopted to address these challenges, most existing solutions rely on basic keyword-matching algorithms or static templates, which restrict their ability to understand context, intent, or complex natural language inputs within unstructured legal text. Such systems often fail to identify nuanced "unfair" clauses, such as unlimited indemnification or hidden non-compete scopes, leading to irrelevant or inaccurate risk assessments that undermine user trust. Furthermore, many general-purpose AI tools lack domain-specific training on legal corpora and are not designed to translate "legalese" into layman's terms effectively.

As a result, there is a critical need for an intelligent, context-aware system that leverages **Natural Language Understanding (NLU)** and **Explainable AI (XAI)** to accurately interpret contract queries, provide real-time risk scoring, and offer simplified explanations. Addressing this gap is essential to enhance the contractual literacy of unrepresented parties, reduce the risk of legal exploitation, and contribute to a more equitable and transparent business environment.

1.4 AIMS AND OBJECTIVES

1.4.1 Aim

The aim of this study is to design and develop an **AI-Powered Contract Clause Risk Analyzer** using Natural Language Understanding (NLU) and Explainable AI (XAI) to help unrepresented parties identify, understand, and mitigate legal risks in commercial agreements.

1.4.2 Objectives

To achieve the aim of the study, the following objectives have been set:

1. To examine the limitations of current manual and keyword-based contract review processes for freelancers and SMEs.
2. To identify the functional and technical requirements for building an NLU-based legal risk detection system using the **Contract Understanding Atticus Dataset (CUAD)**.
3. To develop a prototype system that utilizes **Legal-BERT** for clause classification and Large Language Models (LLMs) for simplifying complex legal prose
4. To evaluate the system's performance in terms of classification accuracy, risk scoring reliability, and the clarity of generated layman explanations.

1.4.3 Research Questions

1. What are the primary challenges unrepresented parties face when reviewing commercial contracts without legal counsel?
2. What specific technical requirements are necessary to enable an AI to accurately classify "high-risk" legal clauses?
3. How can Explainable AI (XAI) techniques be applied to translate "legalese" into accurate layman’s terms?
4. How effective is the developed tool in improving the user’s understanding of contractual risks compared to traditional reading?

1.5 SIGNIFICANCE OF THE STUDY

The rapid digitization of professional services has created an urgent need for scalable and intelligent systems that can enhance legal literacy and streamline contract review for those without formal legal training. This study is significant as it addresses the growing demands for accessible "LegalTech" by exploring the application of **Natural Language Understanding (NLU)** and **Explainable AI (XAI)** in the development of a risk detection system.

The proposed AI-powered analyzer has the potential to transform how unrepresented parties—such as freelancers and startup founders—manage legal obligations by offering 24/7 automated review, reducing the financial burden of legal fees, and ensuring accurate identification of predatory clauses. Unlike traditional keyword searches or basic automated tools, this system leverages **Legal-BERT** and **LLMs** to understand the specific context and intent behind complex legal prose, enabling more meaningful and layman-friendly interactions.

This study contributes to the body of knowledge in the fields of **artificial intelligence**, **human-computer interaction**, and **educational technology** by demonstrating how high-stakes legal data can be simplified without losing accuracy. For small business owners and freelancers, the research offers practical insights into how AI can be strategically adopted to enhance operational safety and contractual confidence. Furthermore, for the broader academic community, the findings can inform future designs of intelligent conversational or analytical systems in other complex domains where expert-level knowledge must be translated for general users.

1.6 JUSTIFICATION OF THE STUDY

As the professional landscape continues to digitize, the demand for scalable, efficient, and accessible legal support systems has become increasingly important. Traditional legal channels, such as hiring private attorneys for contract review, are often overburdened and prohibitively expensive, resulting in unrepresented parties signing agreements with significant delays or total lack of comprehension. This creates a clear need for innovative solutions that can enhance the quality of legal review without overextending the limited financial resources of freelancers and SMEs.

The justification for this study lies in the potential of **Natural Language Understanding (NLU)** and **Explainable AI (XAI)** to address these challenges by offering personalized, context-aware, and real-time risk analysis. Unlike rule-based systems that rely on static keywords, the use of specialized models like **Legal-BERT** enables the system to comprehend and respond to diverse and unstructured legal text more effectively, thus improving the quality of the interaction and the safety of the user.

By leveraging this technology, unrepresented parties can improve their contractual engagement, reduce the risk of signing predatory terms, and ensure consistent delivery of legal information. This research is timely and relevant, as many sectors are exploring AI integration but lack systems tailored specifically for the high-stakes legal needs of non-lawyers. The study fills this gap by not only developing a prototype analyzer but also evaluating its effectiveness in translating complex law into plain English, providing a foundation for further innovation in student-centered or freelancer-centered AI applications. The findings will be beneficial to the legal-tech industry, policymakers, and individual users looking to improve digital equity through intelligent automation.

1.7 LIMITATIONS AND DELIMITATIONS

1.7.1 Limitations

1. **Technical Scope**: The system primarily focuses on **Natural Language Understanding (NLU)** for text-based analysis. Features such as optical character recognition (OCR) for handwritten documents, audio-based legal consultation, or multi-language support will not be included due to time and technical constraints.
2. **Evaluation Scope**: The performance of the risk analyzer will be assessed through a limited number of user simulations and tests. A large-scale deployment across multiple legal jurisdictions is outside the scope of this research.
3. **Data Constraints**: While the system utilizes the **CUAD dataset**, the knowledge base is limited to commercial and service-level agreements. Limited access to proprietary or highly specialized corporate legal data may affect the system's realism in niche legal fields.
4. **Generalizability**: The tool is designed for general commercial use for freelancers and SMEs. Therefore, it may not perfectly reflect the specific regulatory requirements of highly specialized industries like healthcare or criminal law.

1.7.2 Delimitations

To maintain focus and feasibility, the study sets the following boundaries:

1. **Target Audience**: The system is developed specifically for unrepresented parties, including **freelancers, small business owners, and startup founders**. It will not include features tailored for professional attorneys or large corporate legal departments.
2. **Scope of Support**: The system will assist with identifying risks in **Service Agreements, NDAs, and general commercial contracts**. It will not address criminal, family, or real estate law.
3. **Language and Medium**: The application will operate in **English** and support only text-based uploads via a web interface.
4. **Development Tools**: Open-source tools and libraries—specifically **Legal-BERT, Python, and the CUAD dataset**—will be used for implementation, excluding paid proprietary legal software.

1.8 LITERATURE REVIEW

The development of an **AI-Powered Contract Clause Risk Analyzer** sits at the intersection of several rapidly evolving fields: Natural Language Processing (NLP), Explainable AI (XAI), and Legal Technology (LegalTech). The following sections review the current academic landscape relevant to this study.

1.8.1 The "Access to Justice" Gap and LegalTech for SMEs

Literature consistently highlights a "justice gap" where individuals and small-to-medium enterprises (SMEs) are priced out of professional legal services. As noted by **Simshaw (2022)**, the high cost of legal counsel creates a two-tiered system where only large entities can afford comprehensive contract reviews. Recent research into "Justice Tech" suggests that AI can democratize access to legal information by providing 24/7 automated assistance, helping users solve legal problems independently or identifying when professional intervention is strictly necessary (HiiL, 2025).

1.8.2 NLP and Clause Classification using Legal-BERT

Traditional contract analysis relied on keyword searches, which often failed to grasp the semantic nuances of "legalese." The shift toward **Transformer-based models** has revolutionized this domain. **Chalkidis et al. (2020)** introduced **Legal-BERT**, a version of the BERT architecture pre-trained on diverse legal corpora (court cases, contracts, and legislation). Studies show that models fine-tuned on legal-specific data significantly outperform general-purpose models in tasks like clause classification. For instance, **LegalPro-BERT** has demonstrated high accuracy in identifying specific provisions within SEC filings by learning the syntactic structures unique to legal writing (Moonlight, 2025).

1.8.3 The Role of the CUAD Dataset

The **Contract Understanding Atticus Dataset (CUAD)** represents a milestone in open-source legal AI research. Comprising over 13,000 expert-labeled clauses across 510 commercial contracts, it provides the "academic proof" needed to train reliable models. Research utilizing CUAD focuses on 41 categories of important clauses, such as indemnification, change of control, and non-competes, which are critical for corporate transactions and risk assessment (The Atticus Project, 2021).

1.8.4 Explainable AI (XAI) and Layman Simplification

A primary challenge in legal AI is the "black-box" nature of deep learning. In a high-stakes domain like law, transparency is paramount. **Explainable AI (XAI)** aims to provide human-interpretable reasons for AI-generated outcomes. **Kesari & Sele (2025)** argue for a "Legal-XAI" framework that bridges the gap between computer science and law, ensuring that automated decisions are democratic and inclusive. Furthermore, project **CLAUDETTE** has shown success in using machine learning to not only detect "unfair" clauses in online terms of service but also to categorize them in ways that reflect consumer protection standards (Lippi et al., 2019).

1.9 RESEARCH METHODOLOGY

This study adopts the **Design Science Research (DSR)** methodology, which focuses on the creation and evaluation of innovative artifacts to solve identified practical problems. In this case, the artifact is the **AI-Powered Contract Clause Risk Analyzer**. The methodology is structured into five key phases:

1.9.1 Requirement Analysis and Data Collection

The first phase involves identifying the specific needs of freelancers and SMEs through a review of common "unfair" terms in service agreements. The primary data source for training and validation is the **Contract Understanding Atticus Dataset (CUAD)**. This dataset provides the necessary 13,000+ expert-labeled clauses across 41 categories, serving as the academic and legal foundation for the model.

1.9.2 System Architecture and Design

The system will be designed as a web-based application. The architecture consists of three main layers:

- **The Sieve (Classification Layer):** Utilizing **Legal-BERT**, the system will scan uploaded PDF/text documents to segment and classify paragraphs into legal categories (e.g., Indemnification, Limitation of Liability).

- **The Judge (Risk Scoring Layer):** A logic-based engine will evaluate the classified clauses against predefined risk parameters (e.g., flagging "unlimited" or "indefinite" terms as High Risk).

- **The Translator (Simplification Layer):** An integration with a Large Language Model (LLM) such as **GPT-4 or Llama 3** will be used to generate plain-English explanations and suggest neutral alternatives.

1.9.3 Implementation

The development will be carried out using **Python** as the primary programming language. **Hugging Face Transformers** library will be used to implement Legal-BERT, while the web interface will be built using a framework like **Flask or FastAPI**. The LLM component will be accessed via API with strict prompting to ensure legal accuracy is maintained during the simplification process.

1.9.4 Evaluation and Testing

The prototype will be evaluated using two primary methods:

- **Technical Performance:** Measuring the accuracy, precision, and recall of the Legal-BERT model in correctly identifying clauses from a test subset of the CUAD dataset.
- **User Evaluation:** A group of selected participants (freelancers and small business owners) will interact with the tool. They will provide feedback on the clarity of the "layman" explanations and the perceived helpfulness of the risk scores compared to manual review.

1.9.5 Ethical Considerations

To ensure the responsible use of AI in a legal context, the system will include clear disclaimers stating that the output is "Legal Information, not Legal Advice." No personally identifiable information (PII) from the uploaded contracts will be stored, and data privacy will be maintained through temporary session processing.

1.10 RESEARCH ETHICS

This research will adhere to established ethical standards to ensure the integrity of the study and the protection of all participants, particularly given the high-stakes nature of legal document analysis. The researcher is committed to maintaining transparency, objectivity, and respect throughout the research process.

Informed Consent: All participants involved in the evaluation phase will be fully informed about the purpose of the study, which focuses on the development of an **AI-Powered Contract Clause Risk Analyzer**. Participation will be entirely voluntary, and participants will have the right to withdraw at any stage without penalty.

Confidentiality and Data Privacy: Confidentiality will be strictly upheld. No personally identifiable information (PII) from uploaded contracts will be collected or disclosed, and all data gathered for evaluation will be used solely for academic purposes. Technical measures will be implemented to secure data and protect it from unauthorized access.

Legal Disclaimer and Integrity: The researcher will ensure that all referenced work, including the use of the **CUAD dataset** and **Legal-BERT**, is properly cited. Crucially, the system will include a prominent disclaimer stating that the AI's output constitutes **legal information and not professional legal advice**, ensuring the tool is not misused in unintended ways.

Compliance: This research will be conducted in full compliance with the ethical guidelines of the host institution and relevant academic standards for AI and legal technology research.

1.11 CONCLUSION

This chapter has provided a comprehensive overview of the proposed study, detailing the development of an **AI-Powered Contract Clause Risk Analyzer** designed to bridge the "Access to Justice" gap for unrepresented parties. By aligning the research with the requirements of the project proposal, the study has established a clear roadmap for addressing the financial and technical barriers that freelancers and SMEs face during contract review.

The motivation for this research stems from the real-world problem where non-experts often "sign blind" due to the high costs of legal counsel. This chapter introduced the application of **Explainable AI (XAI)** and **Natural Language Processing (NLP)**, specifically leveraging **Legal-BERT** and the **CUAD dataset**, to create a system that not only identifies high-risk clauses but also simplifies them into layman's terms.

The preliminary methodology and literature review support the feasibility of this project, highlighting the shift from traditional keyword-matching to context-aware, intelligent automation. By providing a scalable, low-cost tool for identifying "unfair" terms, this research aims to enhance contractual literacy and promote digital equity. The subsequent chapters will build upon this foundation, providing a detailed review of related literature and a technical breakdown of the system's design and evaluation.

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Appendix A: Sample Questionnaire For Evaluation

Note: This questionnaire uses a 5-point Likert Scale (1 = Strongly Disagree, 5 = Strongly Agree).

Section 1: Perceived Usefulness

1. The AI Analyzer helped me understand complex legal terms I wouldn't have understood otherwise.
2. The risk scores (High/Medium/Low) helped me prioritize which parts of the contract to negotiate.
3. I would feel more confident signing a contract after using this tool.

Section 2: Clarity and Accuracy 4. The "Layman's Translation" provided by the AI was easy to read and follow. 5. The suggested alternative clauses seemed fair and balanced. 6. The interface was intuitive and required little to no training to use.

Section 3: Trust and Ethics 7. I trust the AI's classification of clauses into risk categories. 8. The legal disclaimer clearly explained that this is not a substitute for professional legal advice.

Appendix B: Mock Interaction (Prototype Walkthrough)

User Action: Uploads Freelance_Agreement.pdf

System Processing:

Step 1: Scanning document... (Legal-BERT identifies 14 clauses).

Step 2: Risk analysis complete.

System Output:

ALERT: High-Risk Clause Detected (Indemnification)

Original Text: *"The Contractor shall indemnify and hold harmless the Client from any and all claims, damages, and expenses, including attorney fees, arising out of any act or omission of the Contractor..."*

Risk Score: HIGH

Plain English Translation: You are agreeing to pay for all of the client's legal fees and losses if anything goes wrong, even if it wasn't entirely your fault.

Suggested Alternative: *"Each party shall be responsible for its own acts and omissions. Liability shall be limited to the total value of the contract."*

Appendix C: Consent Form Template

CONSENT FORM FOR PARTICIPATION IN ACADEMIC RESEARCH STUDY

TITLE OF STUDY: Development of an AI-Powered Contract Clause Risk Analyzer

RESEARCHERS: Mathias Yao George Kpodo, Isaac Yeboah, Akogo Paul Akpene

INSTITUTION: Ghana Communication Technology University

Purpose: This study evaluates an AI tool designed to help non-experts identify and understand risky terms in legal service agreements.

Involvement: You will be asked to upload a sample contract (or use a provided one) and complete a 10-minute feedback questionnaire regarding the tool's performance.

Confidentiality: No personal documents uploaded will be stored on our servers. All feedback is anonymized.

Voluntary Participation: You may withdraw from the test at any time.

Consent Statement: By participating in the evaluation, I acknowledge that I have read the above information and voluntarily agree to participate.

Participants Signature: Mathias, Isaac, Paul

Date: 11th December, 2025